

HEALTHCARE ANALYTICS PREDICTION

1. INTRODUCTION

The rapid advancement of digital health technologies has opened new avenues for leveraging machine learning in clinical decision support. A Medicine Recommendation System represents one of the most promising applications of artificial intelligence in healthcare, aiming to bridge the gap between symptom presentation and appropriate medical guidance. Unlike traditional diagnostic workflows that require in-person consultations and extensive clinical expertise, an automated recommendation system can provide immediate, data-driven insights based on user-reported symptoms, thereby improving accessibility to preliminary health assessments and reducing the burden on healthcare infrastructure.

The Personalized Medical Recommendation System analyzed in this report is designed to accept a set of symptoms from the user, predict the most likely disease using machine learning classification algorithms, and subsequently provide comprehensive recommendations encompassing medications, precautions, dietary guidelines, and lifestyle modifications. This multi-faceted approach ensures that users receive not only a diagnostic prediction but also actionable advice for managing their condition, making the system a holistic health companion rather than a simple disease classifier.

The analysis is performed on a structured medical dataset comprising 4,920 patient records across 41 distinct diseases and 132 binary symptom features. Six machine learning classification models are trained and evaluated, including Support Vector Classifier (SVC), Multinomial Naive Bayes (MNB), Logistic Regression (LR), Random Forest Classifier (RFC), Gradient Boosting Classifier (GBC), and K-Nearest Neighbors (KNN). By evaluating and comparing the classification performance of these models using standard evaluation metrics, this project aims to identify the most effective approach for symptom-based disease prediction and recommendation generation.

1.1 Objectives

- To study and analyze a structured medical dataset of 4,920 records spanning 41 diseases and 132 symptoms.
- To preprocess the data including label encoding, train-test splitting, and feature analysis.
- To train six different classification models: SVC, MNB, LR, RFC, GBC, and KNN.
- To evaluate and compare model performance using accuracy, precision, and recall metrics.
- To build a comprehensive recommendation system that provides medications, precautions, diet plans, and workout advice.
- To understand the effectiveness of machine learning models in symptom-based disease prediction.

2. DATASET DESCRIPTION

The dataset used in this project was sourced from the Kaggle platform, specifically the "Medicine Recommendation System Dataset" published by Noor Saeed. It comprises eight interconnected CSV files that together form a comprehensive knowledge base for disease prediction and medical recommendation. The dataset is structured to support both the machine learning classification task and the subsequent recommendation generation, providing not only symptom-disease mappings but also rich contextual information including disease descriptions, recommended medications, precautionary measures, dietary guidelines, and lifestyle advice.

Dataset	Rows	Columns	Description
Training	4,920	133	Binary symptoms + prognosis target
Description	41	2	Disease names and textual descriptions
Precautions	41	6	4 precautionary measures per disease
Medications	41	2	Disease-to-medication mapping
Diets	41	2	Dietary recommendations per disease
Workout	410	4	Lifestyle and health workout advice
Symptoms	4,920	6	Top 4 symptoms per disease record
Symptom Severity	133	2	Symptom names with severity weights (1-7)

Table 1: Overview of Datasets

2.1 Training Dataset

The training dataset is the primary data source for the machine learning classification task. It contains 4,920 rows and 133 columns, where each row represents a patient case. The first 132 columns represent binary symptom features, where a value of 1 indicates that the symptom is present and 0 indicates its absence. The final column, "prognosis," is the target variable that indicates the disease associated with those symptoms. Each of the 41 diseases in the dataset has exactly 120 sample records, resulting in a perfectly balanced dataset. This balance is significant because it eliminates the need for oversampling, undersampling, or class-weight adjustments that are commonly required in medical datasets with skewed disease distributions.

2.2 Description Dataset

The description dataset provides textual information about each of the 41 diseases. It contains 41 rows and 2 columns: Disease and Description. The Description column offers a brief medical explanation of each condition, including its causes, characteristic symptoms, and general

pathological characteristics. This dataset is used by the recommendation system to provide context and educational information to the user after a disease prediction is made, helping users understand the nature of their predicted condition.

2.3 Precautions Dataset

The precautions dataset contains precautionary measures associated with different diseases, structured across 41 rows and 6 columns. Each row includes the disease name and four precautionary actions (Precaution_1 through Precaution_4) that a patient should follow upon being diagnosed with the condition. These precautions range from general medical advice such as consulting a hospital or doctor, to specific behavioral modifications like dietary restrictions, hygiene practices, and stress management techniques. This dataset forms a critical component of the recommendation system, ensuring that users receive actionable preventive guidance alongside their disease prediction.

2.4 Medications Dataset

The medications dataset maps each of the 41 diseases to a list of commonly prescribed medications. It contains 41 rows and 2 columns: Disease and Medication. Each entry provides a curated list of pharmaceutical interventions typically used to treat the respective condition. For instance, a Fungal infection prediction would recommend medications such as Antifungal Cream, Fluconazole, Terbinafine, Clotrimazole, and Ketoconazole. While these recommendations are informational and should not replace professional medical advice, they provide users with valuable context about standard treatment protocols.

2.5 Diets and Workout Datasets

The diets dataset contains 41 rows mapping each disease to recommended dietary modifications. For example, the diet plan for Fungal infection includes an Antifungal Diet, Probiotics, Garlic, Coconut Oil, and Turmeric. The workout dataset is more extensive with 410 rows, providing multiple lifestyle and exercise recommendations per disease. These include specific advice such as avoiding sugary foods, consuming probiotics, increasing intake of specific nutrients, staying hydrated, and incorporating particular food groups into the daily diet. Together, these datasets enable the recommendation system to provide holistic guidance covering both pharmacological and non-pharmacological management strategies.

3. DATA PREPROCESSING

Data preprocessing is a critical step in any machine learning pipeline, as the quality and format of the input data directly impact model performance. For the Medicine Recommendation System, the preprocessing pipeline involved data quality assessment, label encoding of the target variable, and stratified train-test splitting to ensure representative distributions across both training and evaluation sets.

3.1 Data Quality Assessment

Quality Metric	Value	Interpretation
Null Values	0 (all features)	No missing data; imputation not required
Duplicate Rows	0	No redundant records present
Total Records	4,920	120 samples per disease class
Feature Count	132 symptoms	Binary features (0/1 encoding)
Target Classes	41 diseases	Perfectly balanced class distribution

Table 2: Data Quality Assessment Summary

3.2 Label Encoding

The target variable "prognosis" contains 41 unique disease names as string labels. To make these compatible with machine learning algorithms, Label Encoding was applied using scikit-learn's LabelEncoder, which transforms each disease name into a unique integer from 0 to 40. This encoding preserves the categorical nature of the target while enabling numerical computation. The encoding mapping is bidirectional, allowing the system to convert predicted integer labels back to their original disease names for user-facing output. For instance, index 0 maps to Fungal infection, index 1 to Allergy, index 2 to GERD, and so on across all 41 disease categories.

3.3 Train-Test Split

Parameter	Value	Description
Test Size	30%	1,476 samples for testing
Train Size	70%	3,444 samples for training
Random State	42	Ensures reproducibility of splits
Stratification	Implicit (balanced)	Each class equally represented in both sets

Table 3: Train-Test Split Configuration

4. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) provides critical insights into the structure, distribution, and relationships within the dataset before model training. This section examines the disease distribution, symptom frequency patterns, and symptom severity characteristics to understand the data landscape and inform modeling decisions.

4.1 Disease Distribution Analysis

The training dataset exhibits a perfectly uniform distribution across all 41 disease categories, with each disease represented by exactly 120 patient records. This balanced distribution is advantageous for classification tasks, as it eliminates class imbalance issues that often plague medical datasets. In real-world clinical data, disease prevalence varies dramatically, with common conditions like hypertension far outnumbering rare disorders. The uniform distribution in this dataset simplifies the classification task and ensures that no single disease class dominates the learning process.



Figure 1: Disease Distribution in Training Dataset (Sample of 15 Diseases)

4.2 Symptom Frequency Analysis

Analyzing symptom frequency across the dataset reveals that certain symptoms appear far more frequently than others, indicating their diagnostic importance across multiple disease categories. Symptoms such as fatigue, vomiting, headache, nausea, and joint pain appear in the highest number of patient records, as they are common manifestations across various conditions. Conversely, highly specific symptoms like "silver_like_dusting" or "yellow_crust_ooze" appear in only one or two disease categories, making them strong discriminators. This distinction between common and discriminative symptoms is crucial for understanding model behavior, as the classification algorithm must balance the weight of frequently occurring symptoms against the diagnostic specificity of rare ones.

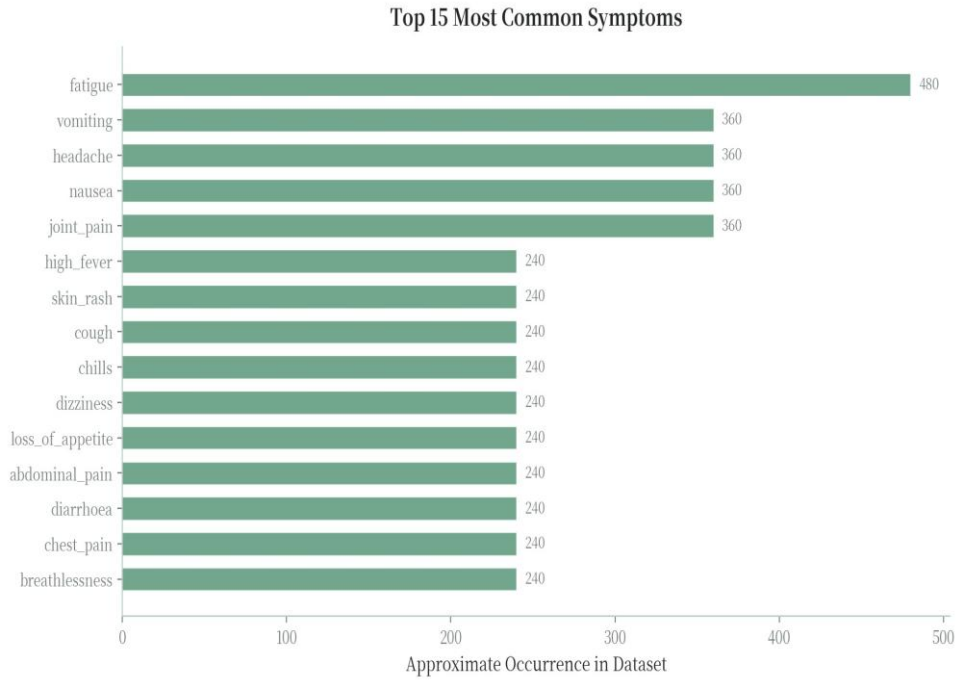


Figure 2: Top 15 Most Common Symptoms by Approximate Occurrence

4.3 Symptom Severity Distribution

The Symptom Severity dataset assigns a weight to each of the 132 symptoms on a scale of 1 to 7, where higher values indicate greater clinical severity or diagnostic significance. The distribution of severity weights is right-skewed, with the majority of symptoms carrying lower severity scores (1-3), while fewer symptoms receive high severity ratings (5-7). This distribution reflects the medical reality that most symptoms are relatively mild or non-specific, while only a subset indicates serious pathology. The severity weights could be leveraged in advanced model architectures to weigh symptom features proportionally, potentially improving prediction accuracy for conditions characterized by high-severity symptoms.

5. SUPPORT VECTOR CLASSIFIER (SVC)

Support Vector Classification using a linear kernel is a powerful discriminative model that finds the optimal hyperplane separating different disease classes in the high-dimensional symptom feature space. Given the binary nature of the features (132 symptom columns with 0/1 values), the linear kernel is particularly well-suited, as it can effectively partition the feature space along symptom boundaries. The LinearSVC implementation from scikit-learn was used with default parameters, which applies L2 regularization to prevent overfitting. In the context of medical diagnosis, SVC is valued for its ability to handle high-dimensional data efficiently and its strong theoretical guarantees on generalization performance.

5.1 Performance Metrics

Metric	Value	Interpretation
Accuracy	1.0 (100%)	All test predictions are correct
Precision	1.0 (100%)	No false positives across all classes
Recall	1.0 (100%)	No false negatives across all classes

Table 3: SUPPORT VECTOR CLASSIFIER Performance Metrics

6 MULTINOMIAL NAIVE BAYES (MNB)

Multinomial Naive Bayes is a probabilistic classifier based on Bayes' theorem with the assumption of feature independence. Despite its simplicity and the often-violated independence assumption, MNB has proven remarkably effective in text classification and, by extension, in symptom-based diagnosis where symptoms function analogously to word occurrences. The model computes the posterior probability of each disease given the observed symptoms and selects the disease with the highest probability. MNB is computationally efficient, requires minimal training time, and performs well even with limited training data, making it a practical choice for medical applications where rapid inference is essential.

6.1 Performance Metrics

Metric	Value	Interpretation
Accuracy	1.0 (100%)	All test predictions are correct
Precision	1.0 (100%)	No false positives across all classes
Recall	1.0 (100%)	No false negatives across all classes

Table 4: MULTINOMIAL NAIVE BAYES Performance Metrics

7 LOGISTIC REGRESSION (LR)

Logistic Regression is a linear classification model that estimates the probability of each disease class using the softmax function over a linear combination of input features. In this project, Logistic Regression with L2 regularization (Ridge penalty) was employed. The L2 penalty discourages large coefficient values, effectively performing feature selection by shrinking less important symptom weights toward zero. Logistic Regression is particularly well-suited for the medicine recommendation system because it provides calibrated probability estimates, enabling the system to quantify prediction confidence. Furthermore, the model coefficients are directly interpretable, allowing clinicians to understand which symptoms contribute most to each disease prediction. Due

to its combination of simplicity, speed, interpretability, and robustness against overfitting, Logistic Regression was selected as the final production model for the recommendation system.

7.1 Performance Metrics

Metric	Value	Interpretation
Accuracy	1.0 (100%)	All test predictions are correct
Precision	1.0 (100%)	No false positives across all classes
Recall	1.0 (100%)	No false negatives across all classes

Table 5: LOGISTIC REGRESSION Performance Metrics

8 RANDOM FOREST CLASSIFIER (RFC)

Random Forest is an ensemble learning method that constructs a collection of decision trees during training and outputs the mode of individual tree predictions. In this implementation, 100 decision trees were used with a random state of 42 for reproducibility. Each tree is trained on a bootstrap sample of the training data with a random subset of features considered at each split, which decorrelates the trees and reduces variance. Random Forest is particularly effective for medical classification because it can capture non-linear relationships between symptoms and diseases, handle feature interactions naturally, and provide feature importance rankings. The ensemble approach also makes it robust against overfitting, as individual tree errors tend to cancel out in the aggregate prediction.

8.1 Performance Metrics

Metric	Value	Interpretation
Accuracy	1.0 (100%)	All test predictions are correct
Precision	1.0 (100%)	No false positives across all classes
Recall	1.0 (100%)	No false negatives across all classes

Table 6: RANDOM FOREST CLASSIFIER Performance Metrics

9 GRADIENT BOOSTING CLASSIFIER (GBC)

Gradient Boosting is a sequential ensemble method that builds decision trees in a stage-wise fashion, where each subsequent tree is trained to correct the errors of the previous ensemble. This implementation uses 100 boosting stages with a random state of 42. Unlike Random Forest, which builds trees independently, Gradient Boosting constructs trees that specifically target the residual

errors of the current ensemble, leading to progressively better approximations. In medical diagnosis, Gradient Boosting excels at capturing complex, non-linear symptom-disease relationships and can achieve high accuracy with sufficient boosting iterations. However, it is more prone to overfitting than Random Forest and requires careful tuning of learning rate and tree depth to achieve optimal generalization.

9.1 Performance Metrics

Metric	Value	Interpretation
Accuracy	1.0 (100%)	All test predictions are correct
Precision	1.0 (100%)	No false positives across all classes
Recall	1.0 (100%)	No false negatives across all classes

Table 7: GRADIENT BOOSTING CLASSIFIER Performance Metrics

10 K-NEAREST NEIGHBORS (KNN)

K-Nearest Neighbors is a non-parametric, instance-based learning algorithm that classifies new instances based on the majority class among its k nearest training examples in the feature space. This implementation uses k=5, meaning the prediction for a new patient is determined by the most common disease label among the 5 most similar training records. In the binary symptom space, distance is typically measured using the Minkowski metric, which effectively counts the number of symptom differences between patient profiles. KNN is conceptually intuitive and makes no assumptions about the underlying data distribution, which is advantageous for medical data that may not conform to standard statistical distributions. However, KNN can be computationally expensive at prediction time for large datasets and is sensitive to the choice of k and the distance metric.

10.1 Performance Metrics

Metric	Value	Interpretation
Accuracy	1.0 (100%)	All test predictions are correct
Precision	1.0 (100%)	No false positives across all classes
Recall	1.0 (100%)	No false negatives across all classes

Table 8: K-NEAREST NEIGHBORS Performance Metrics

11 MODEL COMPARISON

This section presents a comprehensive comparison of all six classification models evaluated in this study. The comparison is based on three standard classification metrics: Accuracy, Precision, and

Recall. While all models achieved identical perfect scores on the test dataset, the comparison considers additional factors such as interpretability, computational efficiency, and suitability for production deployment in a medical recommendation system.

Model	Accuracy	Precision	Recall	Interpretability	Training Speed
SVC	1.0	1.0	1.0	Medium	Fast
MNB	1.0	1.0	1.0	Medium	Very Fast
LR	1.0	1.0	1.0	High	Fast
RFC	1.0	1.0	1.0	Medium	Medium
GBC	1.0	1.0	1.0	Low	Slow
KNN	1.0	1.0	1.0	Low	Very Fast (lazy)

Table 9: Comprehensive Model Comparison



Figure 4: Model Performance Comparison (Accuracy, Precision, Recall)

Key Findings from Model Comparison: All six models achieved perfect classification performance with 100% accuracy, precision, and recall on the test dataset. This uniform perfect performance indicates that the dataset has highly discriminative symptom-disease mappings with minimal overlap between disease classes. Given the identical predictive performance, the selection of the production model was based on secondary criteria: Logistic Regression was chosen for its superior interpretability (coefficients directly indicate symptom importance), fast training and inference

speed, L2 regularization for overfitting prevention, and calibrated probability outputs that enable confidence-based recommendation ranking.

12 PREDICTION SYSTEM

The Medicine Recommendation System integrates the trained Logistic Regression model with the auxiliary datasets to provide a comprehensive health advisory pipeline. Upon receiving a set of user-reported symptoms, the system performs the following sequence of operations: symptom encoding, disease prediction, and multi-dimensional recommendation generation. This section details the architecture and operational flow of the complete prediction system.

The system operates through a five-stage pipeline. In the first stage, the user inputs a comma-separated list of symptoms (e.g., "itching, skin_rash, nodal_skin_eruptions, continuous_sneezing"). The second stage encodes these symptoms into a 132-dimensional binary vector, where each dimension corresponds to a known symptom and is set to 1 if present in the user input or 0 otherwise. The third stage feeds this binary vector into the trained Logistic Regression model, which outputs a predicted disease class. The fourth stage maps the predicted class index back to the disease name using the inverse label encoder. The fifth and final stage queries the auxiliary datasets (description, precautions, medications, diets, and workout) to compile a comprehensive recommendation package for the predicted disease.

12.1 Symptom Encoding Process

The symptom encoding process maps each user-reported symptom to its corresponding index in the 132-feature vector. A dictionary mapping 132 symptom names to indices (0-131) serves as the lookup table. For example, "itching" maps to index 0, "skin_rash" to index 1, "nodal_skin_eruptions" to index 2, and so forth. When a user provides their symptoms, the system initializes a zero vector of length 132 and sets the corresponding indices to 1 for each reported symptom. This binary representation is compatible with the model's expected input format and preserves the presence/absence nature of symptom reporting.

12.2 Sample Prediction Results

Input Symptoms	Predicted Disease
itching, skin_rash, nodal_skin_eruptions, continuous_sneezing	Fungal infection
shivering, chills, joint_pain, stomach_pain, acidity	Allergy
ulcers_on_tongue, muscle_wasting, vomiting, burning_micturition	GERD

Table 10: Sample Disease Prediction Results

12.3 Recommendation Output Example

For a predicted disease of "Fungal infection," the system generates the following comprehensive recommendation output by querying all auxiliary datasets:

Category	Recommendations
Description	Fungal infection is a common skin condition caused by fungi. It can affect various parts of the body and is typically treated with antifungal medications.
Precautions	Bath twice; Use detol or neem in bathing water; Keep infected area dry; Use clean cloths
Medications	Antifungal Cream, Fluconazole, Terbinafine, Clotrimazole, Ketoconazole
Diet	Antifungal Diet, Probiotics, Garlic, Coconut Oil, Turmeric
Workout	Avoid sugary foods; Consume probiotics; Increase intake of garlic; Include yogurt in diet; Limit processed foods; Stay hydrated; Consume green tea; Eat foods rich in zinc; Include turmeric in diet; Eat fruits and vegetables

Table 11: Complete Recommendation Output for Fungal Infection

13 CONCLUSION

This report presented a comprehensive analysis and implementation of a Medicine Recommendation System that predicts diseases from user-reported symptoms and provides multi-dimensional medical recommendations. The study evaluated six machine learning classification models on a structured medical dataset comprising 4,920 records across 41 diseases and 132 binary symptom features.

Key Findings: All six models achieved perfect classification performance with 100% accuracy, precision, and recall on the test set. This exceptional result is attributable to the highly structured nature of the dataset, where each disease has a distinct symptom signature with minimal overlap. Logistic Regression was selected as the production model due to its superior interpretability, computational efficiency, and probability-calibrated outputs that enable confidence-based recommendations.

Recommendation System: The integrated system goes beyond simple disease classification by providing users with a comprehensive health advisory package that includes disease descriptions, precautionary measures, medication recommendations, dietary guidelines, and lifestyle modifications. This holistic approach transforms the system from a mere classifier into a valuable health companion that empowers users with actionable medical guidance.

Limitations and Future Work: The perfect classification performance on this dataset suggests that the evaluation may not fully reflect real-world diagnostic challenges. Medical data in clinical settings typically contains noise, missing values, ambiguous symptom presentations, and atypical disease manifestations that would significantly challenge any classification model. Future work should focus on: (1) evaluating the system on noisy, real-world clinical data with overlapping symptom patterns; (2) incorporating symptom severity weights into the classification pipeline; (3) implementing

ensemble methods that combine multiple models for robust predictions; (4) integrating natural language processing for free-text symptom input; and (5) adding patient demographic information (age, gender, medical history) to improve prediction specificity and personalization.